



SIMATS Online Programming Lab

DS PROGRAMMING



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QUESTIONS

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AVL Tree- Insert Elements

AVL TREE- INSERT ELEMENTS

"Write a program to create an AVL Tree and insert the following elements:
50, 30, 70, 20, 40, 60, 80. Construct the AVL Tree, Display the tree using Inorder Traversal,
Verify that the height balance property is maintained after each insertion."

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 struct Node {
4     int key;
5     struct Node *left, *right;
6     int height;
7 };
8 int height(struct Node *n) {
9     return n ? n->height : 0;
10 }
11 int max(int a, int b) {
12     return (a > b) ? a : b;
13 }
14 struct Node* newNode(int key) {
15     struct Node* node = (struct Node*)malloc(sizeof(struct Node))
16     node->key = key;
17     node->left = node->right = NULL;
18     node->height = 1;
19     return node;
20 }
21 struct Node* rightRotate(struct Node* v) {
```

OUTPUT

Inorder traversal of AVL tree: 20 30 40 50 60 70 80

INPUT

Your INPUT goes here!

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